

Module 3 – Debugging Exercise on Pandas Code LAB ASSIGNMENT

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Instructions: Identify the error(s) in each code snippet and correct the code

- Write a brief reason for the correct code

Q1. Data Loading Error

Error: `pd.readcsv()` is incorrect function name

Correct Code:

```
import pandas as pd
df = pd.read_csv('data.csv')
```

Explanation:

The correct function is `read_csv()` (with underscore), not `readcsv`. It is used to load CSV files into a DataFrame.

Q2. DataFrame Creation Issue

Error: Incorrect dictionary syntax (`'Age': 20, 21`)

Correct Code:

```
data = {'Name': ['Aman', 'Riya'], 'Age': [20, 21]}
df = pd.DataFrame(data)
```

Explanation:

Each key must have a list of values. Here, Age should be a list `[20, 21]`, not separate values.

Q3. Indexing Mistake

Error: Missing closing bracket]

Correct Code:

```
df = pd.read_csv('data.csv')  
df.loc[1:5, 'Salary']
```

Explanation:

.loc[] requires proper syntax with both row and column selection inside brackets.

Q4. Filtering Error

Error: Incorrect placement of condition inside brackets

Correct Code:

```
df[df['Salary'] > 50000]
```

Explanation:

The condition must be written as df['column'] > value inside the DataFrame filtering brackets.

Q5. Group-by Syntax Error

Error: Using square brackets instead of parentheses

Correct Code:

```
df.groupby('Department').mean()
```

Explanation:

groupby is a function, so it must be called using parentheses (), not [].

Q6. Missing Function Call

Error: Missing parentheses in function call

Correct Code:

```
df.drop_duplicates()
```

Explanation:

Functions must be called using (). `drop_duplicates()` removes duplicate rows from a DataFrame.